

- Outline how the genetic engineering process produced Bt maize plants with resistance to insect attack.

(b) A proposed conservation benefit of Bt maize is that, unlike chemical insecticide sprays, it should not harm non-target insects.

In 1999, a laboratory experiment tested the effects of feeding milkweed leaves treated in different ways to the caterpillars of monarch butterflies. Each group of caterpillars was provided with milkweed leaves to eat throughout the experiment. The three different treatments were:

- Fig. 4.1 shows the results of this laboratory experiment.

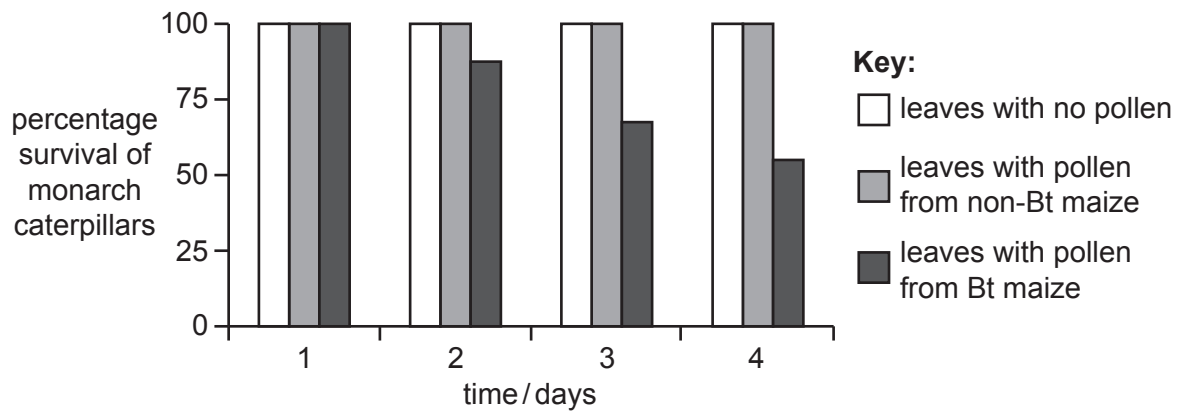


Fig. 4.1

(i) State what can be concluded from Fig. 4.1.

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(ii) Laboratory results are not always useful in predicting what will happen in real ecosystems.

- Some scientists predicted that increasing the area of Bt maize in the USA would decrease the number of monarch butterflies, based on the results in Fig. 4.1.
- More monarch butterflies were counted in 2019 than in 1999, even though the quantity of Bt maize grown more than tripled.

Suggest reasons why the predicted decrease in the number of monarch butterflies did not occur in the real ecosystem.

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